ENHANCING CYTOLOGICAL STAINING-FREE IMAGE CLASSIFICATION WITH ROBUST VISION TRANSFORMERS

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Abstract The Vision Transformer (ViT) architecture has emerged as a transformative approach in computer vision, offering scalability and global attention, garnering significant interest in recent years. Its flexibility has encouraged exploration across diverse domains, including the demanding field of medical imaging, where datasets are often explicitly tailored to complex tasks, such as those in small sample sizes. In this work, we investigate the boundaries of the DEIT III architecture and introduce a suite of combined mixed regularisation and augmentation techniques to improve model performance using a novel loss function and a smoothly differentiable activation function designed to stabilise training and yield better performance. Our approach, designed to classify cell cytologies, was tested for stratification of bladder cancer invasiveness using staining-free cell images. It was further benchmarked against several medical imaging datasets and compared against strong baselines, including ViT, ConvNeXtV2, and ResNet-50. Experimental results show that our proposed model consistently outperforms, demonstrating exceptional capabilities in image classification and advancing the state of the art in vision transformer applications for complex medical imaging tasks.

1. INTRODUCTION

Deep learning techniques have revolutionised medical image analysis, with a significant impact on cytological cell segmentation tasks, such as the detection of bladder cancer from urine samples. Convolutional Neural Networks (CNN) have historically dominated the field, offering high performance in both classification. However, their reliance on local receptive fields often limits their ability to capture long-range dependencies. The introduction of ViT [1] offers a compelling alternative by incorporating global self-attention mechanisms that capture both local details and holistic image context. Urine cytology for bladder cancer diagnosis presents unique challenges, such as high morphological variability in cell shapes, sizes, and textures; severe class imbalance; and the limited number and heterogeneity of datasets, which increase the risk of overfitting.

To address these challenges, we propose a DIstillation with NO labels (DINO)-Data efficiency image Transformer (DeiT)-based classification model (Fig. 1) that integrates multiple architectural and training enhancements. Our approach incorporates the Log-Softplus ERror Activation Function (SERF) activation function [2] into the attention modules, replacing the conventional softmax, thereby enhancing the model's ability to focus on subtle cellular features by improving the sharpness and sparsity of attention weights for discrimination between overlapping sparse feature structures.

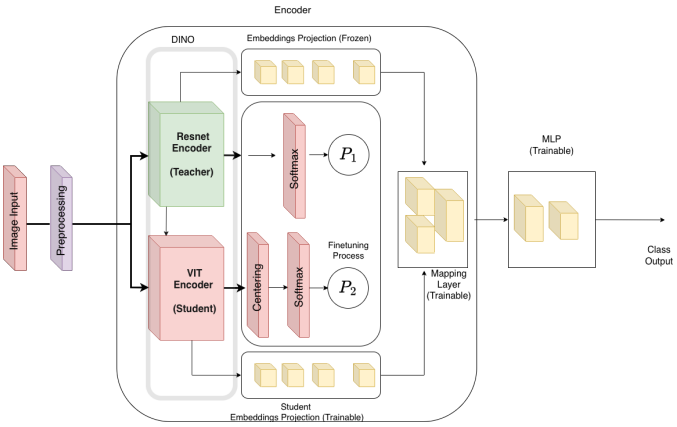


Fig. 1: Overview of DINO with ViT: The student network mimics the teacher output using two augmented views.

The proposed architecture uses a patch embedding process aligned with ViT in a teacher-student setup via the DINO scheme. Cell images from imaging flow cytometry (Cytek Amnis ImageStreamX) are split into fixed-size patches, linearly projected into embeddings, and augmented with positional encodings. Multiple transformer encoder layers, enhanced with SERF-based multi-head attention, process these embeddings. Class imbalance is addressed using Asymmetric Loss function (ASL), which downweights easy negatives and emphasizes rare positives, while data augmentations (image normalization and patch-level mixing) improve robustness to acquisition variations and spatial distortions.

The model is evaluated on public datasets and an in-house bladder cancer cell dataset (CELLo), performing multiclass

classification of tumour stage (T stage). Comparisons include standard ViT [1] with imbalance modifications, ConvNeXtV2 [3], and a modified ResNet-50 [4].

The key contributions of this work are:

- Incorporation of SERF activation to stabilise gradients and mitigate vanishing/exploding effects.
- Integration of Asymmetric Loss to address severe class imbalance;
- Comparison study across different datasets and model benchmarks to assess the performance of the proposed model.

The remainder of this paper is organised as follows: Section 2 presents a summary of related works, Section 3 details the employed methodology, Section 4 presents and discusses the obtained results, concludes the paper and suggests future work directions.

2. LITERATURE REVIEW

ViTs has significantly reshaped the field of computer vision by adapting the Transformer architecture originally designed for Natural Language Processing (NLP) tasks into image domains, proposing both pure ViTs architectures and hybrid approaches combining convolutional operations and self-attention mechanisms. These innovations aim to enhance model explainability, data efficiency, and stabilise the training of deep networks. Notably, DeiTIII [5] builds directly on the original ViT [1], introducing strategies such as 3-Augment, layer scaling [6], and stochastic depth [7] regularisation to improve model robustness and performance.

In the field of urinary cytology, automated bladder cell classification has emerged as a critical research focus due to its clinical relevance in diagnosis and monitoring. Early approaches primarily employed CNN. Moriya et al. [8] developed a deep learning–based recognition model for an automated slide evaluation system, reporting robust performance across common urinary cell types. To address dataset scarcity. Lee et al. [9] investigated transfer learning by pretraining CNNs on lung cell datasets before fine-tuning on urinary cells, significantly improving classification accuracy and generalisation. Data augmentation strategies such as CutMix [10] and MixUp [11] have also been shown to enhance model stability and mitigate overfitting in small cytology datasets [8].

Recently, ViT-based frameworks have emerged. For example, [12] introduced a weakly supervised CrossViT architecture composed of two feature paths combined with attention in intermediate layers, initially aimed at classifying whole-slide images (WSIs), achieving 84.8% accuracy and an AUC of 85.4%, outperforming traditional CNN baselines. CellVTA [13] integrated convolutional adapters within a ViT backbone to better capture fine-grained spatial details of small cellular structures, improving segmentation and classification in complex cytology images. Similarly, CytoFM [14] proposed a ViT-based foundation model pretrained in a self-supervised fashion on large-scale cytology datasets, demonstrating superior generalisation for downstream urinary cell classification compared to models pretrained on natural or histopathology images. In parallel, large-scale efforts, such as [15] and the VISIOCYT-1 clinical study [16, 17], have validated the potential of AI for urinary cytology, reporting diagnostic sensitivity substantially higher than that of conventional cytology. Beyond cytology slides, ViTs have also been successfully applied to bladder endoscopic imagery, achieving higher accuracy than traditional CNN pipelines [18], thereby demonstrating their adaptability across urology-related imaging modalities.

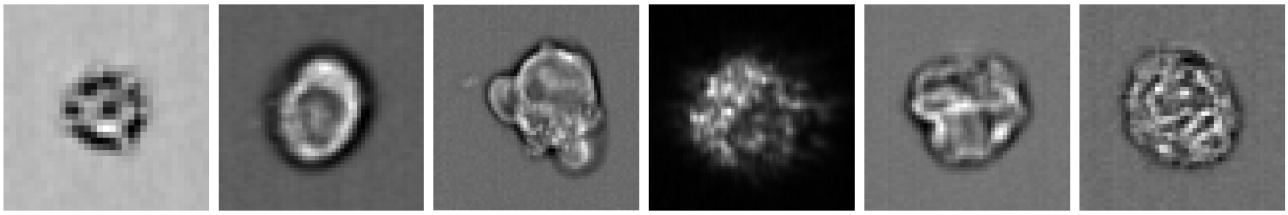
Overall, the literature suggests that CNN-based architectures remain the traditional image backbone for research on bladdercell classification, with transfer learning and augmentation serving as key enablers for robust performance on smaller datasets. However, the emergence of ViT-based frameworks demonstrates clear potential for improving both classification accuracy and interpretability in urinary cytology via ViT-based models. However, these ViT approaches still pose challenges, including long convergence/training times, the need for large training datasets, and severe class imbalance, motivating further architectural modifications and tailored regularisation strategies to improve performance, convergence and stability during training.

3. METHODOLOGY AND BACKGROUND

To evaluate our DINO-DeiT modified model, we benchmarked it on multiple medical imaging datasets against established architectures, including ViT-Base [1], modified ResNet50 [4], and ConvNeXtV2 [19]. All models were compared using modifications to loss and activation functions to address class imbalance, enabling a clear assessment of convolutional versus transformer-based approaches, including self-supervised and pretraining-augmented variants.

3.1. CELLo Dataset Description

The CELLo project aims to investigate a non-invasive, complementary approach to cystoscopy for the early and personalised diagnosis of bladder cancer, based on urine cellular images in a staining-free fashion. Given the disease's high recurrence rate (up to 70%) and the limitations of current invasive cystoscopy and often inconclusive cytology methods (which miss 20-40% of bladder cancers and ¿40% of low-grade tumours), CELLo focuses on isolating and concentrating cancer cells from urine. This non-invasive strategy seeks to enable staining-free, precise, and patient-specific diagnostics, advancing the field of personalised cancer care. The images (Fig. 2) were acquired using imaging flow cytometry under brightfield microscopy with standardised acquisition settings to ensure reproducibility.



(a) na (b) ta (c) t1 (d) t2 (e) t3 (f) t4

Fig. 2: Labelled single-cell images. The first row shows the raw samples; the second row shows the corresponding layer labels.

Captured at a total magnification of 60×, the cell images are provided in a greyscale brightfield spectrum. This heterogeneity requires models to effectively handle noisy, low-contrast images while ensuring consistent performance when extracting representative features. High-grade tumours are characterised by highly pleomorphic cells that exhibit aggressive growth and carry a significantly higher risk of invasion and metastasis (Figure 3). Patient groups are categorised into groups encompassing multiple fields of view or time points, enabling patient-level analysis and preventing data leakage during model training.

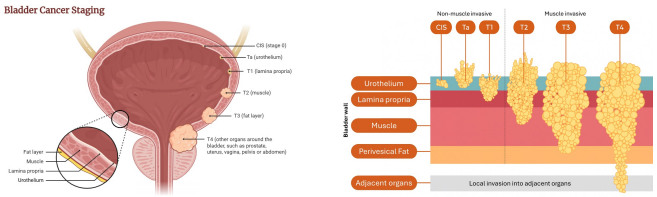


Fig. 3: (a) Section of bladder tumour stages representation. (b) 2D overview of the classification.

Single-cell images are linked to patient-level metadata with diagnostic labels (healthy vs. cancerous). Cancer samples include tumour stage and grade: stages range from non-muscle-invasive (Ta–T1) to muscle-invasive (T2–T4), while grades are low-grade (LG) or high-grade (HG); "Na" denotes normal cases.

3.2. Teacher Student - DINO

DINO is a self-supervised learning framework where a **student network** learns to mimic the output of a **teacher network** in a self-supervised way (Figure 1). When combined with ViT- based models, DINO leverages the transformer's ability to model long-range dependencies in images while learning rich feature representations. Considering a student network g_s and a teacher network g_t, both based on ViT, we define a centre vector $C \in \mathbb{R}^K$. The student and teacher networks are controlled by temperatures t_{ps} and t_{pt}, respectively, while the teacher network and the centre vector are updated using momentum rates l and m. Given a batch of images $x \in \mathbb{R}^{n\times H\times W\times 3}$, DINO applies two random augmentations to generate views x_1, x_2:

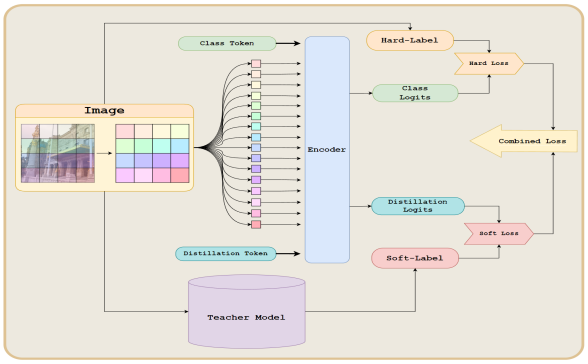


Fig. 4: Depiction of DeiT's distillation token.

$$x_1, x_2 = \text{augment}(x) \tag{1}$$

The student and teacher outputs are then:

$$s_1 = g_s(x_1), \quad s_2 = g_s(x_2), \quad t_1 = g_t(x_1), \quad t_2 = g_t(x_2) \tag{2}$$

The cross-entropy loss between teacher and student, with centring and sharpening, is defined as:

$$s = \text{softmax}\left(\frac{s}{t_{ps}}\right) \tag{3}$$

$$t = \text{softmax}\left(\frac{t - C}{t_{pt}}\right) \tag{4}$$

$$H(t, s) = -\frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K} t_{ik} \log s_{ik} \tag{5}$$

$$\mathcal{L} = \frac{1}{2}H(t_1, s_2) + \frac{1}{2}H(t_2, s_1) \tag{6}$$

The network parameters are updated as follows. The student network is optimised using standard gradient descent on the loss $\mathcal{L}$:

$$\theta_s \leftarrow \theta_s - \eta\nabla_{\theta_s}\mathcal{L} \tag{7}$$

The teacher network is updated using a momentum-based rule:

$$\theta_t \leftarrow l\,\theta_t + (1 - l)\,\theta_s \tag{8}$$

Finally, the centre vector C is updated based on the teacher outputs:

$$C \leftarrow m\,C + (1-m)\frac{1}{2n}\sum_{i=1}^{n}(t_{1,i} + t_{2,i}) \tag{9}$$

The selected ViT backbone is based on the DeiT-III framework [5] and comprises three main components. Patch embedding divides an image into fixed-size patches and projects them into d-dimensional vectors, also known as visual tokens. The transformer encoder applies multi-head self-attention layers with residual connections to capture long-range dependencies, and the MLP head projects the [CLS] token output to K dimensions, aligning with the teacher's output. Combined with DINO via output mapping layer, this architecture provides a robust self-supervised learning framework, where momentum teacher updates, centring, and temperature scaling enable the model to learn robust feature representations without labels.

3.3. DEIT-III teacher backbone modification

The primary focus of the backbone teacher model will be on the DeiTIII architecture (Fig. 4), complemented by the DINO scheme. It will be compared with other State-of-the-art (SotA) network models. This architecture will be probed at its **S**mall version for a more concise exploration of results, and the sets described are used solely to fine-tune the available pre-trained weights of this architecture from the HuggingFace [20].

The DeiT model is a variant of the ViT that utilises the ViT 's parallelism across tokens to achieve the two objectives simultaneously. The model's loss is calculated at two different endpoints of the network: one at the `[class]` token, where a simple Cross-Entropy can be used for classification via *hard-labels*, and one at the *distillation* token, where the Kullback-Leibler divergence (KL) [21] between the *soft-labels* of the two models is minimised. The total loss of the network is calculated by combining the results of each loss through a balancing λ and a temperature τ factors (Equ. 10) where ψ is the softmax [22] function, and Z_n corresponds to the logits of network n.

$$\begin{aligned}\mathcal{L}_{total} = &(1-\lambda)\mathcal{L}_{\text{Hard}}(\psi(Z_{student}), y) \\ &+ \lambda\tau^2\mathcal{L}_{\text{Soft}}(\psi(Z_{student}/\tau), \psi(Z_{teacher}/\tau))\end{aligned} \tag{10}$$

The implementation involves adapting and transferring model weights within the PyTorch framework to incorporate a modified attention mechanism. The process follows four main steps:

1. **Exporting Pre-Trained Weights:** Pre-trained weights from the HuggingFace repository were exported for subsequent reload into the modified architecture.
2. **Modifying the Attention Mechanism:** The original attention module was replaced with a modified version. Only the final activation function was replaced while preserving the structural compatibility of input/output tensor shapes, parameter names, and layer structure.
3. **Transferring Weights to the Modified Model:** The exported weights were loaded into the modified architecture. Consistent structure and parameter naming enabled weight loading without additional transformations.
4. **SERF Integration:** To improve training stability and mitigate gradient vanishing, we incorporate Hybrid Smooth Operators. Specifically, the SERF activation function and regularisation methods such as asymmetric loss and stochastic depth are integrated into DeiT-III. This ASL variant is designed to enhance performance under high class imbalance.

The model is fine-tuned on medical datasets using a DINO scheme, and its ability to transfer generalised representations to cytology tasks is evaluated. This approach enables integrating a custom attention mechanism while retaining knowledge embedded in the pre-trained weights.

3.4. Baseline Models

For comparison purposes, three additional model architectures (ResNet-50, ConvNextV2, and Vit + SERF) are also benchmarked across all evaluated datasets to assess their potential and serve as a measure of trustworthiness for the different models and corresponding modifications.

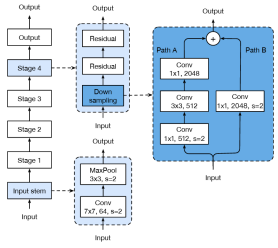
3.4.1. ResNet-50

As a strong convolutional baseline, the original ResNet-50 [4] provides hierarchical feature extraction via residual learning. However, the original architecture employs an initial 7×7 convolution with a stride of 2, followed by max pooling, resulting in a substantial downsampling factor of 32 in the early convolution stages. This aggressive reduction may discard critical, small, fine-grained cellular features in medical imaging tasks [23] [24]. To mitigate this limitation, we adapt the stem of ResNet-50 by replacing the first convolution with a stack of smaller 3×3 kernels and reducing the downsampling rate (Fig. 5), thereby preserving higher spatial resolution.

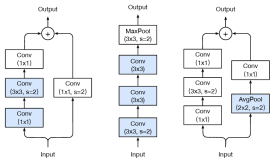
3.4.2. ConvNeXtV2

ConvNextV2 [19] is a modern CNN that incorporates design ideas from Vision Transformers while maintaining efficient convolutional operations. Model key innovations include the use of fully fused inverted bottlenecks to reduce the memory footprint and Global Response Normalisation (GRN), which stabilises convergence during training and provides deep feature representations (Figure 6).

The network is composed of hierarchical stages similar to ResNets, each containing several *ConvNeXt blocks*



Original ResNet-50 stem with 7×7 convolution and stride 2



Modified ResNet: smaller 3×3 convolutions, shifted downsampling, average pooling

Fig. 5: Comparison of original and modified ResNet-50 stems to preserve spatial detail in early layers.

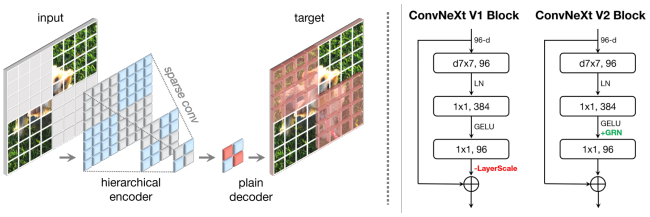


Fig. 6: ConvNeXtV2 block and hierarchical stage structure.

with a residual connection. For an input feature map $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, a single block performs:

$$\mathbf{Y} = \mathbf{X} + \mathcal{F}(\mathbf{X}), \quad \mathcal{F}(\mathbf{X}) = \sigma\Big(\text{DWConv}(\text{LN}(\mathbf{X}))\Big) \odot \text{PWConv}_2(\text{PWConv}_1(\mathbf{X})), \tag{11}$$

where LN is Layer Normalisation, DWConv is depthwise convolution, PWConv_1 and PWConv_2 are pointwise 1×1 convolutions, σ is a non-linear activation, and $\odot$ denotes optional gating.

Each stage reduces spatial resolution via the corresponding strided convolutions,

$$\mathbf{X}^{(s+1)} = \text{Downsample}(\mathbf{X}^{(s)}), \quad s = 1, \ldots, S-1, \tag{12}$$

halving the given input resolution while doubling the number of channels. A global average pooling and an MLP are placed on the final stage to produce classification outputs:

$$\mathbf{y} = \text{Softmax}(\mathbf{W} \cdot \text{GAP}(\mathbf{X}^{(S)}) + \mathbf{b}). \tag{13}$$

These architectural refinements enable ConvNeXtV2 to preserve fine-grained spatial features. By combining the efficiency of CNNs with a transformer-inspired design, providing a strong baseline for high-variability imaging tasks.

3.4.3. Vision Transformer with SERF Activation.

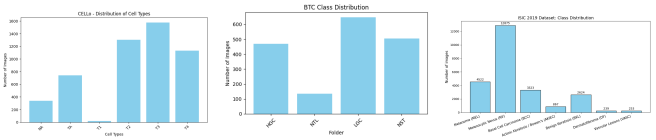
The baseline ViT follows the standard architecture where te input image is divided into non-overlapping patches, linearly projected to token embeddings, and processed by a stack of Transformer encoder blocks. Each block consists of multi-head self-attention and a feed-forward network (FFN), both wrapped in Layer Normalisation and residual connections. In the original ViT configuration, the FFN layer uses the GELU activation function, that can lead to gradient saturation and suboptimal curvature in deep architectures, causing training problems. To improve optimisation stability, we replace GELU in all FFN layers with the SERF activation. SERF is defined as

$$\text{SERF}(x) = x \cdot \tanh(\log(1 + e^x)),$$

and provides smoother gradients stability. Since SERF is fully compatible with the standard FFN layer, is used as a drop-in replacement for GELU without modifying the overall ViT architecture, while preserving structural simplicity.

3.5. Benchmarking and Evaluation

To ensure comparability and model robustness, we benchmark all models across multiple datasets. The CELLo dataset, our target user case dataset, the International Skin Imaging Collaboration (ISIC) 2019 [25], consists of a large-scale collection of dermoscopic images for classification, containing 25,331 training images, 8,000 validation images, and 1,512 test images across seven disease categories. with imbalance and high intra-class variability. Additionally, we use the Bladder Tissue Endoscopics Classification (BTC) dataset [26], which includes multi-domain endoscopic images for bladder tissue classification. Both datasets serve as a complementary benchmark outside image cytology, exhibiting notable distributional characteristics and class imbalances (Fig. 7).



(a) CELLo dataset class distribution (b) BTC dataset class distribution (c) ISIC 2019 dataset class distribution

Fig. 7: Class distributions across the three datasets used in our experiments.

4. EXPERIMENTS AND RESULTS

For classification, we evaluate our proposed DINO-DeiT model pretrained with self-supervised DINO and fine-tuned on the CELLo dataset. The remainder of the benchmark model architectures include ResNet-50, which offers a strong, simple convolutional baseline for comparison; ConvNeXtV2, which leverages modernised convolutional blocks with improved efficiency and representation learning; and a modified ViT that includes SERF, capable of capturing long-range dependencies in complex image structures.

Table 1: Performance of models on CELLo, ISIC 2019, and BTC datasets.

Data	Model	AUC	W-Acc	Acc
CELLo	DINO-DEiT	0.92	0.89	0.90
	ResNet50	0.87	0.85	0.86
	ViT+SERF	0.83	0.80	0.82
	ConvNeXtV2	0.79	0.77	0.78
ISIC	DINO-DEiT	0.86	0.84	0.85
	ResNet50	0.84	0.82	0.83
	ViT+SERF	0.82	0.80	0.81
	ConvNeXtV2	0.81	0.79	0.80
BTC	DINO-DEiT	0.88	0.85	0.86
	ResNet50	0.83	0.81	0.82
	ViT+SERF	0.81	0.79	0.80
	ConvNeXtV2	0.80	0.78	0.79

Each model is optimised using a consistent data augmentation pipeline that includes random rotations, horizontal and vertical flipping, intensity jitter (±10%). These transformations increase robustness to staining variability, a common issue in urine-derived cytological images. All models were trained for 150 epochs, with early stopping applied when convergence was rapid. The learning rate was selected using a cosine annealing schedule, improving robustness and generalisation. Stratified split on all datasets using a 70-10-20 partition (Training, Validation, Test). Table 1 summarises the performance of all evaluated models across the CELLo, ISIC 2019 (skin lesions), and BTC datasets. Metrics reported include AUC-ROC, balanced accuracy, and accuracy.

Figures 8–10–9 provide a comprehensive comparison of model performance across different datasets. Each figure presents the evaluation of four models (ViT+SERF, ResNet50, DINO-DEiT Modified, and ConvNeXtV2) across three key aspects: train and validation loss convergence, AUC curves, and confusion matrices on the test set. Rows correspond to the models evaluated, while columns correspond to the metrics, enabling a clear visual comparison of training stability, classification performance, and error patterns.

This layout facilitates the assessment of the effectiveness of self-supervised pretraining (DINO-DEiT) relative to standard supervised backbones, as well as the impact of architectural modifications, such as SERF activation, on ViT.

The results show that the proposed DINO-DeiTIII model consistently outperforms ResNet-50 across all datasets, achieving superior tissue-level classification performance. This gain stems from the ability of DINO-ViT-based architectures to model long-range dependencies, capture inter-class variability, and leverage self-supervised pretraining effectively.

ResNet-50 performs well on smaller datasets, while ConvNeXtV2 provides an efficient alternative but exhibits training instability that requires careful tuning. In contrast, the vanilla ViT achieves the lowest performance, underscoring the importance of architectural adaptations—such as imbalance-aware losses and data-efficient pretraining—to address severe class imbalance.

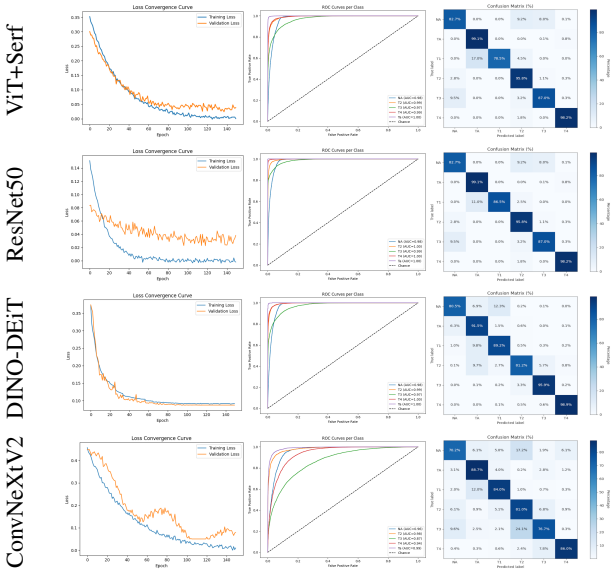


Fig. 8: CELLo dataset: Loss, AUC, and Confusion Matrix for four models.

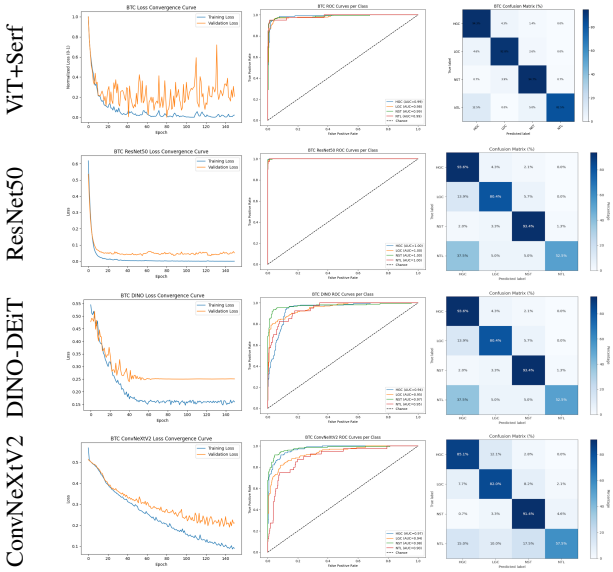


Fig. 9: BTC dataset: Loss, AUC, and Confusion Matrix for four models.

The proposed modification of the DeiTIII teacher combined with the DINO student framework significantly improves accuracy on small, high-resolution images by preserving spatial context and enabling robust fine-grained classification. The inclusion of SERF and the DINO-DeiT configuration further enhances performance under imbalanced conditions, with the largest gains observed on the CELLo dataset, which exhibits the highest imbalance ratio.

In urine cell classification, the proposed model achieves up to a 3.8 p.p. improvement over ConvNeXtV2 and 4.5

p.p. over ResNet-50 on highly imbalanced datasets, while also demonstrating faster convergence and improved stability across cross-validation folds. Attention maps indicate a stronger focus on clinically relevant regions, reducing gradient oscillations during training.

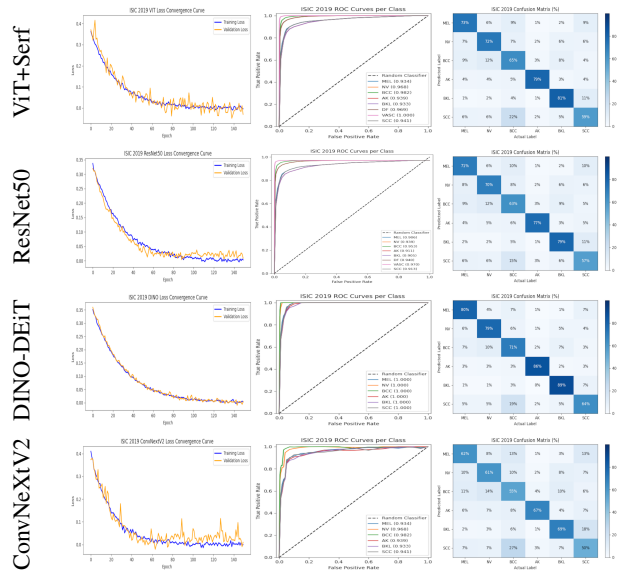


Fig. 10: ISIC 2019 dataset: Loss, AUC, and Confusion Matrix for four models.

Overall, this work demonstrates that transformer-based models, when combined with specialised activation functions, imbalance-aware loss functions, self-supervised learning, and domain-specific augmentations, can significantly advance bladder endoscopy and urine cytology classification in staining-free images. Future work should address domain shift across acquisition protocols and explore additional regularisation strategies to further improve generalisation in settings with extreme morphological variability and limited annotations.

Acknowledgements

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References

[1] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020.

[2] S. Nag, M. Bhattacharyya, A. Mukherjee, and R. Kundu, "Serf: Towards better training of deep neural networks using log-softplus error activation function," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2023, pp. 5324–5333.

[3] S. Woo, S. Debnath, R. Hu, X. Chen, Z. Liu, I. S. Kweon, and S. Xie, "Convnext v2: Co-designing and scaling convnets with masked autoencoders," 2023. [Online]. Available: https://arxiv.org/abs/2301.00808

[4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," 2015. [Online]. Available: https://arxiv.org/abs/1512.03385

[5] H. Touvron, M. Cord, and H. Jégou, "Deit iii: Revenge of the vit." in *ECCV (24)*, ser. Lecture Notes in Computer Science, S. Avidan, G. J. Brostow, M. Cissé, G. M. Farinella, and T. Hassner, Eds., vol. 13684. Springer, 2022, pp. 516–533. [Online]. Available: http://dblp.uni-trier.de/db/conf/eccv/eccv2022-24.html#TouvronCJ22

[6] H. Touvron, M. Cord, A. Sablayrolles, G. Synnaeve, and H. Jégou, "Going deeper with image transformers." *CoRR*, vol. abs/2103.17239, 2021. [Online]. Available: http://dblp.uni-trier.de/db/journals/corr/corr2103.html#abs-2103-17239

[7] G. Huang, Y. Sun, Z. Liu, D. Sedra, and K. Weinberger, "Deep networks with stochastic depth," 2016, cite arxiv:1603.09382Comment: first two authors contributed equally. [Online]. Available: http://arxiv.org/abs/1603.09382

[8] H. Moriya, H. Ohkubo, T. Shimizu, K. Suzuki, K. Ohe, and M. Gotoh, "Urine cell image recognition using a deep-learning model for an automated slide evaluation system," *Cytopathology*, vol. 32, no. 6, pp. 814–821, 2021.

[9] S. Lee, H. J. Ryu, Y. M. Kim, and C.-Y. Park, "Automated classification of urinary cells: Using convolutional neural network pre-trained on lung cells," *Applied Sciences*, vol. 13, no. 3, p. 1763, 2023.

[10] S. Yun, D. Han, S. J. Oh, S. Chun, J. Choe, and Y. Yoo, "Cutmix: Regularization strategy to train strong classifiers with localizable features," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 6023–6032.

[11] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," *arXiv preprint arXiv:1710.09412*, 2017.

[12] Y. Zhao, W. Shen, Y. Fan, H. Zhu *et al.*, "Less: Label-efficient semi-supervised learning for urinary cytology whole slide images," *arXiv preprint arXiv:2306.03407*, 2023.

[13] Y. Fan, Y. Zhao, X. Sun, Y. Wang *et al.*, "Cellvta: Vision transformer adapter for generalizable cell segmentation and classification," *arXiv preprint arXiv:2504.00784*, 2024.

[14] Y. Zhao, Y. Fan, X. Sun *et al.*, "Cytofm: Cytology foundation model for generalizable cell classification," *arXiv preprint arXiv:2504.13402*, 2024.

[15] M. Tsuneki *et al.*, "Development of a deep learning model to assist the diagnosis of neoplastic and nonneoplastic urine cytology whole slide images," *Diagnostics*, vol. 12, no. 1, p. 122, 2022.

[16] J.-B. Roche, A. Chevrot *et al.*, "Performance of the visiocyt® test for the diagnosis of bladder cancer: a prospective multicenter study," *World Journal of Urology*, vol. 39, pp. 1571–1578, 2021.

[17] J. Roche, A. Chevrot *et al.*, "Artificial intelligence for the detection of urothelial carcinoma in urine cytology: Visiocyt® performance in a multicentric prospective study," *European Urology Supplements*, vol. 20, no. 1, p. e441, 2021.

[18] J. Gu, D. Yang *et al.*, "Vision transformer for bladder cancer detection in endoscopic images," *Urologic Oncology*, 2024, in press.

[19] S. Woo, S. Debnath, R. Hu, X. Chen, Z. Liu, I. S. Kweon, and S. Xie, "Convnext v2: Co-designing and scaling convnets with masked autoencoders," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 16 133–16 142.

[20] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, J. Davison, S. Shleifer, P. von Platen, C. Ma, Y. Jernite, J. Plu, C. Xu, T. L. Scao, S. Gugger, M. Drame, Q. Lhoest, and A. M. Rush, "Transformers: State-of-the-art natural language processing." in *EMNLP (Demos)*, Q. Liu and D. Schlangen, Eds. Association for Computational Linguistics, 2020, pp. 38–45. [Online]. Available: http://dblp.uni-trier.de/db/conf/emnlp/emnlp2020-d.html#WolfDSCDMCRLFDS20

[21] S. Kullback and R. A. Leibler, "On information and sufficiency," *The Annals of Mathematical Statistics*, vol. 22, no. 1, pp. 79–86, 1951.

[22] J. S. Bridle, "Probabilistic interpretation of feedforward classification network outputs, with relationships to statistical pattern recognition," in *Neurocomputing: Algorithms, Architectures, and Applications*. Springer-Verlag, 1989, pp. 227–236.

[23] X. Li, Y. Chen, X. Li, W. Zhao, and J. Yang, "Improving detection of tiny objects by preserving fine-grained features in backbone architectures," in *Proceedings of the 12th International Conference on Pattern Recognition Applications and Methods (ICPRAM 2023)*. SCITEPRESS, 2023, pp. 116–127. [Online]. Available: https://www.scitepress.org/publishedPapers/2023/116435/pdf/index.html

[24] T. He, Z. Zhang, H. Zhang, Z. Zhang, J. Xie, and M. Li, "Bag of tricks for image classification with convolutional neural networks," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 558–567.

[25] N. C. Codella, V. Rotemberg, P. Tschandl, M. E. Celebi, S. W. Dusza, D. Gutman, B. Helba, A. Kalloo, K. Liopyris, M. Marchetti *et al.*, "Skin lesion analysis toward melanoma detection 2018: A challenge hosted by the international skin imaging collaboration (isic)," in *arXiv preprint arXiv:1902.03368*, 2019. [Online]. Available: https://www.kaggle.com/datasets/salviohexia/isic-2019-skin-lesion-images-for-classification

[26] J. F. Lazo, B. Rosa, M. Catellani, M. Fontana, F. A. Mistretta, G. Musi, O. de Cobelli, M. de Mathelin, and E. D. Momi, "Endoscopic bladder tissue classification dataset," 2023, supported by the ATLAS project (EU Horizon 2020, Grant 813782) and French State Funds (ANR-11-LABX-0004, ANR-10-IAHU-02). [Online]. Available: https://doi.org/10.5281/zenodo.7741476